PeriGen, Inc.

PeriWatch Vigilance® 2024.1 Bundle
Supporting Document – Technical Description

Version 2024.1.000

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PeriGen
Advanced
Perinatal Systems

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2024-04

Product

PeriWatch Vigilance 2024.1 Bundle

- **PeriWatch Vigilance 2024.1.000**
- **PeriWatch Vigilance HUB 2024.1.000**
- **PeriWatch Vigilance Cues 2024.1.000**
- **PeriWatch Vigilance Curve 2024.1.000**

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Intended Use

PeriCALM® Tracings™ is intended for use as a central monitoring system with signal analysis and display functions and remote repeaters in the perinatal clinical environment. It interfaces with standard fetal and maternal monitors and provides fetal surveillance, monitoring of labor progress, charting, reporting and data analysis, display and archiving functions.

PeriWatch Vigilance® Cues™ is intended for use as an adjunct to qualified clinical decision-making during antepartum or intrapartum obstetrical monitoring at ≥36 weeks gestation to obtain annotation of the FHR for baseline, accelerations and decelerations.

PeriWatch Vigilance® Curve™ is intended for use as a data analysis system to assess labor progress in singleton, cephalic presentation pregnancies when the gestational age is ≥ 35 weeks.

CONTRAINDICATIONS: None known.

WARNINGS AND PRECAUTIONS:

Evaluation of FHR during labor and patient management decisions should not be based solely on PeriWatch Vigilance® Cues™ annotations.

Federal law restricts this device to sale by or on the order of a physician.

Safety and effectiveness of PeriWatch Vigilance® Cues™ at a gestational age<36 weeks has not been established.

PeriWatch Vigilance® Cues™ is intended to be used as an adjunct to fetal surveillance systems that provide standard alerts when fetal heart rates are out of bounds.

Safety and effectiveness of PeriWatch Vigilance® Curve™ at a gestational age<35 weeks has not been established.

Please refer to the User Guide prior to the first use.
Rx only.

Various aspects of the PeriWatch® software suite are subject to issued and pending patents in several jurisdictions. Issued patents include:

USA	6,907,284
USA	7,113,819
USA	6,423,016
USA	8,636,676 B2
USA	8,870,793
USA	9,439,597
USA	9,805,164 B2
USA	10,134,490

Product Lifetime

Use of this software is limited to the period of time for which Microsoft supports the Windows operating system on which it is installed, as well as other prerequisite components that were installed with the software. Check Microsoft's website for further information on Microsoft support policies. Note: there may be further limitations in your contract with PeriGen, requiring you to regularly update to more recent versions of the software as they become available.

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1 Preface

This document will provide technical supportive information for the customer IT and/or the system admin regarding PeriWatch Vigilance on a Windows Server.

1.1 Product Compatibility

The product compatibilities are outlined in section 3 of this document.

1.2 Product Critical Dependencies

PeriWatch Vigilance is fully relying on data received from PWS.
As such, PWS system is required for proper operation, high availability, and proper performance.

1.3 Who should read this document?

This manual is designed for use by Customer IT and Customer representative that are in a role of system administrator. The system administrator should understand the overall strategy of information systems of the unit and the hospital. They must be available to provide consultation and to implement the process.

1.4 Related Documentation

- VIG 2024.1.000 - DMR 10056 - Vigilance Installation Guide
- VIG 2024.1.000 - DMR 10173 - HUB Admin Installation Guide
- VIG 2024.1.000 - DMR 10174 - System migration guidelines
- VIG 2024.1.000 - DMR 10178 - PeriWatch Vigilance User Guide
- VIG 2024.1.000 - DMR 10176 - Clinical Configuration Workbook
- HUB 2024.1.000 - DMR 10175 - PeriWatch Vigilance HUB Admin Tool User Guide
- VIG 2024.1.000 - DMR 10404 - Entra ID (Azure AD) Configuration Guide

2 Configuration overview

2.1 General

The **PeriWatch Vigilance Single Patient Dashboard** is a web-based screen that displays data for one mother including: the fetal heart rate(s) and uterine pressure tracing analyzed by **PeriWatch Vigilance Cues**, labor progress analyzed by **PeriWatch Vigilance Curve,** as well as a graph of maternal vital signs over time.

PeriWatch Vigilance HUB is a web-based multi-patient view with color coding and filtering functionalities based on configurable thresholds. **PeriWatch Vigilance HUB** is a module of **PeriCALM Tracings**, a computer-based information management system for obstetrical care.

The **PeriWatch Vigilance Single Patient Dashboard** can be launched either from **PeriWatch Vigilance HUB, PeriWatch Surveillance System** or from a third party EMR system.

The **PeriWatch Vigilance Single Patient Dashboard** can post an abnormality indicator next to any factor it assesses. The indicator uses the same thresholds and colors that are used for the corresponding item in **PeriWatch Vigilance HUB**.

2.2 Supported configurations

PWS and Vigilance

PeriWatch Surveillance system used for fetal monitoring (EFM) with Cues, with the addition of Vigilance and HUB.

Standalone Vigilance

Vigilance with a third-party fetal monitoring (EFM) system, PeriWatch Surveillance provides a backend for Vigilance.

3 Main system characteristics

The chapter below outlines the software and hardware requirements for the PeriWatch Vigilance system and the supported software platforms.

Note: This listing is provided as an overview; see the documents listed in section 1.4 for full details of system requirements.

3.1 Supported platforms

Supported Servers Operating System:

- Windows 2016 Standard Edition with IIS 10.x
- Windows 2019 Standard Edition with IIS 10.x
- Windows 2022 Standard Edition with IIS 10.x

Supported Client internet browsers:

- Minimum support browsers: Chrome 103.0.5060.66, Edge 103.0.1264.62 and IE version 11 (not support downgrading the IE version by forcing compatibility mode), We strongly recommend using Chrome as the preferred browser for the Vigilance system.

3.2 Hardware requirements

Minimum hardware requirement per number of monitored beds:

	"All in One" application server (PWS +Vigilance)				**Dedicated CDS server**			
# Of monitored locations	**50**	**100**	**200**	**500**	**50**	**100**	**200**	**500**
# Of vCPUs	4	8	12	12	4	6	8	8
GB RAM	12	16	20	24	8	12	12	16
Disk space [GB]	PWS requirements + Requirement of dedicated CDS server				100			
SQL space [GB] [1]	PWS requirements + Requirement of dedicated CDS server				40	60	120	300

(1) For sites where discoverability type set to 3 (To know more about discoverability feature please refer to section 9 in VIG 2024.1.000 - DMR 10056 - Vigilance Installation Guide); the disk space value for 50 monitored beds is sufficient for a period of 5 years.

- For Test server, use the configuration of 50 monitored beds as the applicable configuration.
- Configuration for sites with more than 500 monitored beds requires review by PeriGen Product Development.

3.3 Software requirements

- .NET version 4.8
- MS SQL 2016 / MS SQL 2019 / MS SQL 2022 - Standard Edition
- .NET 8.0.1 Windows Hosting
- VC++ 2013 & VC++ 2015-2022 Redistributable (self-install by the installation wizard)
- PowerShell version 5 or higher.
- Patterns ports 91, 92, 7802 free on the operating system.
- The following ports are the standard ports for Vigilance: 8081,8082,8083,8084,8085, 8203

3.4 Software Bill-of-Materials

Vigilance is compiled using Microsoft Visual Studio; the SQLite database is used for local data storage.

As with any software, there are multiple other modules used internally for various purposes. A detailed SBOM including all modules and versions is available on request.

3.5 Special requirements\restrictions

- All prerequisites need to be installed beforehand including the fetal monitoring system (PeriWatch Surveillance).
- Windows firewall Service is running.
- Certificate for secured configuration (HTTPS)
  - Certificate must be named for the Fully Qualified Domain Name of the server
  - For secure access to SQL, certificate must have the AT_KEYEXCHANGE set to 1.
- A domain account OR Managed Service Account (MSA)
  - That its password should never expire.
  - With local admin rights and “Log on as batch job” and “Log on as service” permission.
- Requirement for AD:
  - There are a minimum of 2 AD groups that need to be set up for Vigilance.

    - HUBUsers AD group – Used for general User access. A separate AD group is required for each facility (for per-facility authorization);
    - HUBAdmin AD group – Used for Administrative functions in HUB Admin. This group also has default access to Vigilance Reports and general user access.
- The maximum length of the server name should be below 15 characters.
- The maximum length for FQDN should be below 63 characters.
- The maximum length for facility name or locales should be below 60 characters.
- The names for all the above should be limited to the following characters:
    - Alphanumeric with or without dash or underscore only

3.6 Official product version

The official Vigilance system version will be provided by PeriGen technical representatives as per the project timeline; the media with the software package will be kept on the designated Vigilance servers.

3.7 Guidelines and Considerations prior to platform updates/changes

This chapter will provide you with guidelines and consideration prior and after of performing changes or updates to the system platform. Such changes\updates could be hardware or software changes including new software installation next to the Vigilance system.

Hardware

The hardware as it appears in the hardware requirement section was defined to ensure proper and reliable operation of the system, and has been tested to ensure continuous work during load and recoverability.
It is therefore not recommended to decrease any hardware characteristics.

Software
As a general rule, it is recommended to first run any change on the test server a few weeks prior to running it on the production server.

- Anti-virus programs may prevent access to the ports the Patterns system uses., If there are any Anti-virus program changes, ensure the ports are still available for the Vigilance system. See also section 7 for further information on Anti-virus configuration.

- For environments where SQL, SQL Server Reporting Services (SSRS), and Vigilance are installed on the same server, avoid port conflicts between SSRS and Vigilance. SSRS 2019 uses port 8082 by default. This is in conflict with the Vigilance system which also uses port 8082. In order to avoid this issue, a new port should be defined for SSRS following these steps:

    a. Stop Reporting service.

b. Open "C:\Program Files\Microsoft SQL Server Reporting services\SSRS\RSHostingService\config.json" and change 8082 to a different port.

c. Using CMD, execute the following commands:

- netsh http delete urlacl url=http://+:8082/
- netsh http add urlacl url=http://+:*NEWPORT#*/ user="NT SERVICE\SQLServerReportingServices"

d. Reboot the server.

4 SQL DB characteristics

The chapter below outlines the DataBase structure and requirements for the PeriWatch Vigilance system.

Note: This listing is provided as an overview; see the documents listed in section 1.4 for full details of system requirements.

4.1 Supported SQL versions

- MS SQL 2016 Standard Edition
- MS SQL 2019 Standard Edition
- MS SQL 2022 Standard Edition

4.2 SQL accounts

To install the Vigilance system, the installation wizard requires SQL user which has DBA Admin privileges on SQL Server. Another alternative is to build the DB in advanced using scripts that will be executed by DBA Admin of site (For more information about that option, please refer to VIG 2024.1.000 – DMR 10056 Vigilance Installation Guide chapter Appendix - Self-install of Vigilance DBs).

During creation of the Vigilance DB the installer will allow the provided account to have access to the DB:

- Domain service Account - which is provided during the installation (A domain account that its password should never expire, with local admin rights and “Log on as batch job” permission.

4.3 Vigilance DataBase

The following DBs will be created:

HUBArchiveDB_<year> (e.g. HUBArchiverDB_2022)

Archived visits will be moved from the main HUBDB to the relevant yearly archive DB; the DB creation script will create a head 5 archive DBs starting with the current year.

Note: in case and there are no future archive DB; the *DatabasePartitions_MaintainPartitions* SP should be executed to create archive DB ahead.

HUBBulkDB

This volatile DB holds all visits data prior to processing.

HUBDB

This is the main system DB which holds system settings and processes visits data.

4.4 Vigilance Jobs

The following Jobs will be set during DB creation:

HUBDB.Archiver
Job is responsible to move data of archived visits from the main HUBDB to the relevant archived year database.

The 2 jobs below will be set only if DBs backup are not being taken care globally by the site (self-backup):
HUBDB.BackupPartitions
Job is responsible to create backup files of Vigilance DBs.
HUBDB.BackupCleaner
Job is responsible to clear obsolete backup files.

4.5 SQLite

The SQLite database is used for local data storage.

5 System maintenance

In general, the Vigilance system is under maintenance contract, as such PeriGen support is responsible to remedy an issue or concern associated with the system. Moreover, the client executive will advise if a new vigilance release is available and the benefit of upgrading.

5.1 Software maintenance and updates

PeriGen regularly reviews and prioritizes software issues, and will release updates from time to time according to that prioritization. This includes issues reported by clients or found at client sites, as well as issues identified by PeriGen R&D that could affect system performance or cybersecurity. When maintenance updates are released, clients who are potential candidates for the update will be notified as to the availability and content of the version update and invited to schedule a time for installation.

5.2 General maintenance tasks

This section will elaborate on which maintenance activities can be done by the client representative without reaching out to PeriGen support.

1. Monitor disk space on the server side and increase space without changing the disk structure and\or clearing files.
2. Monitor SQL DB disk size consumption and increase space.
3. Installation of Microsoft security patches
4. Renew certificates on the server and on ports as per the guidelines below:

Note:	The following can also be done using PeriGen utility “Certificate Check”; To receive that utility please reach out to PeriGen support.

a. Using CMD execute the following command and keep the result in notepad:
   netsh http show sslcert
b. Remove the default pattern port by running the following command:
   netsh http delete sslcert ipport=0.0.0.0:7802
c. Remove the default Notification port by running the following commands:
   netsh http delete urlacl url=http://+:8203/DBService
   netsh http delete sslcert ipport=0.0.0.0:8203
d. Add the default pattern port again but now with the new certificate by running the following command (can also be done form power shell – netsh enter, https enter – rest of command).
   netsh http add sslcert ipport=0.0.0.0:7802 certhash={thumbrint from certificate} appid={Application random GUID from the notepad in #3 above}

e. Add the notification port now with the new certificate by running the following commands:
   http add urlacl url=https://+:8203/DBService user="NT AUTHORITY\Authenticated Users"
   netsh http add sslcert ipport=0.0.0.0:8203 certhash={ thumbrint from certificate}
   appid={Application random GUID from the notepad in #a above}
f. Do the same for plugin port with the new certificate by running the following command:
   netsh http add sslcert ipport=0.0.0.0:7803 certhash={thumbrint from certificate}
   appid={Application random GUID from the notepad in #3 above}
g. Open IIS manager and bind the certificate to each one of the webServices port (8081,8082,8083,8084,8085).
h. IISReset

5.3 Backup and recovery

The hospital should ensure regular backup of the following directories:

- C:\Program Files (x86)\Perigen
- C:\patterns-data

Recovery from backup should be accompanied by PeriGen technical support, in order to verify that the system is complete and consistent after recovery. In the event of a need to recover from backup, PeriGen support personnel should be advised so that they can provide relevant instructions and accompany the recovery.

6 System Security Options

This chapter will elaborate on which security configuration or consideration are available to the customer through configuration options, software options etc.

Customer is responsible for physical security, e.g. preventing unauthorized physical access to servers, and for ensuring that only authorized administrators can log on to the server operating system.

PeriWatch Vigilance is designed to protect PHI (Protected Health Information) and prevent access to PHI by unauthorized users. As such, user authentication and is required before PHI can be displayed.

6.1 User authorization

User authentication and authorization is required before PHI (Protected Health Information) can be displayed.
PeriWatch Vigilance can use the hospital's Active Directory or Azure AD infrastructure for authentication. please refer to the HUB Admin User Guide for further information on how to achieve that.

6.2 Data encryption

All data in transit (i.e., being sent between computers, e.g., from the server to a workstation for display) is sent encrypted using HTTPS. HTTPS and TLS 1.2 are used to encrypt data in transit; PHI data at rest are encrypted using AES-128.

6.3 Inactivity duration

By default, PeriWatch Vigilance will logout user from the system after predefine time of inactivity:

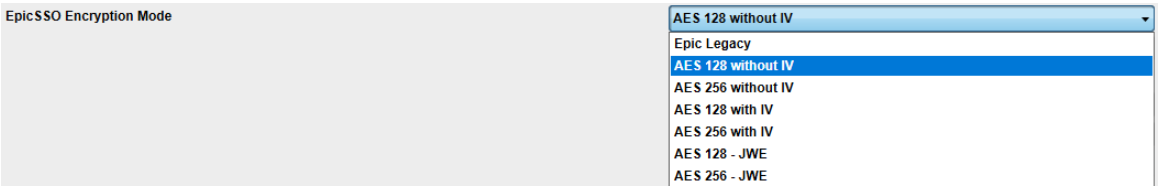
- HUB client - After 60 minutes of inactivity (with the exception of workstations configured as HUB kiosk/Bunker).
- Vigilance client – After 3 minutes of inactivity.

6.4 Single Sign On (SSO)

PeriWatch Vigilance supports Single Sign On integration with PeriWatch Surveillance, so that a user can launch the Vigilance single patient view directly from PWS.

PeriWatch Vigilance supports Single Sign On integration with EHR systems, so that a user can launch the Vigilance single patient view directly from within the EHR system.

The initial implementation of SSO requires configuration by a PeriGen technical expert. In case an SSO encryption mode change is required; open Settings page within the HUB Admin utility and select the required SSO encryption mode.



6.5 Pass-through Authentication

If enabled, pass-through authentication allows automatic authentication to HUB clients as the currently logged-in Windows user, removing the need to separately authenticate on HUB client.

Access to HUB with pass-through authentication is through port 8085 (by default).
i.e., https://<FQDN>:8085

Windows pass-through authentication is enabled by default on the server, but needs to be separately enabled (using domain policy) for the workstations using pass-through authentication.

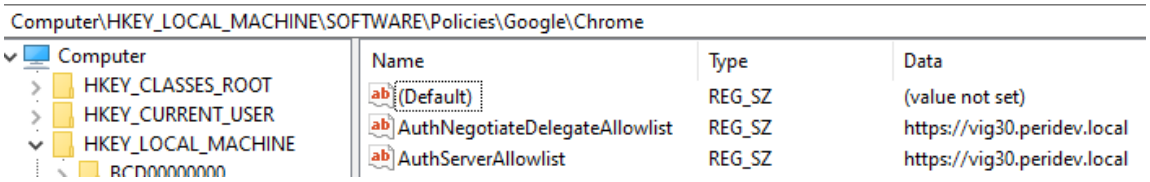
To disable pass-through authentication on the server, stop the Vigilance_WINAUTH webservice via the IIS management console.

To enable pass-through authentication the customer will need to apply the following on the relevant machines using domain policy:

Chrome browser configuration

1. Create the following registry path:
   `Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Policies\Google\Chrome`
2. Create the following string values with the Vigilance server FQDN:
   `AuthNegotiateDelegateAllowlist value` **Error! Hyperlink reference not valid.**
   `AuthServerAllowlist value` **Error! Hyperlink reference not valid.**

   For example:



Edge browser configuration

Enable automatic logon with current username and password.

1. In Internet Explorer, navigate to **Settings > Internet Options**.
2. On the Security tab, add https://<FQDN>:8085 to the Trusted Sites.
3. Under Trusted Sites, click **Custom Level**.
4. Under User Authentication, select **Automatic logon with current user name and password**.

7 Anti-Virus

The hospital is responsible for installing anti-virus software on the Patterns server and client workstations, and for keeping it up-to-date.

7.1 Anti-Virus Exclusions

Antivirus software can degrade performance or interfere with operation of both our client and server software. Real-time scans on our production server can slow down our database access, and scheduled scans can consume dangerous amounts of system resources. Additionally, files can be locked while they are scanned, which can interfere with some parts of the application.
On the server, exclude antivirus scans to follow the directory locations (including all subdirectories).

Note: **Drive Directories may differ due to the location you have installed Vigilance on your Server. The folder names are subject to change.**

- C:\Program Files (x86)\Perigen
- C:\patterns-data

Port scan exclusion, The PeriWatch Vigilance system uses the following ports: 91, 92, 7802, 8081,8082,8083,8084,8085,8203.

It is recommended to disable scanning of these ports, or at most disable blocking.

8 Firewall and network configuration

The hospital is responsible for installing appropriate firewall and keeping it maintained and up-to-date.

Inbound network traffic to the Vigilance server should be limited to the hospital's Local Area Network or Wide Area Network.

8.1 Web proxy and filtering software

ALL Web Proxy/ Filtering Software or Services must be configured to ignore our applications traffic.
Web proxies can incorrectly view our internal communications as internet browsing, and silently route those communications through their proxy server.
This will always impact our application's performance and can result in failure and errors due to blocked or mishandled communication.
Confirm that the web proxy's configuration is set to ignore traffic to the IP addresses and host names of our production servers.

9 Hazard situations

Failure of the IT-network to provide the characteristics and services required for the purpose of the Vigilance system may prevent correct operation of the system but does not result in any hazardous situation directly or indirectly for the system users or patients.

in case of a failure, the system will not be updated with the latest data from the surveillance system or may be unavailable. Many failure scenarios are self-correcting in that the system will automatically restart, recover the data, and attempt to continue, but PeriGen technical support should be contacted in the event of a failure, to identify and correct the problem.

The Vigilance system is intended to be used as an adjunct to fetal surveillance systems that provide standard alerts when fetal heart rates are out of bounds.